01 FOUNDATIONAL CONCEPTS OF ANALYTICS

01 BUSINESS ANALYTICS

- Business analytics is a discipline that includes the use of data, statistical methods, and quantitative tools to solve business problems, identify opportunities and support strategic decisions

Example: A retail chain uses sales data and customer trends to decide which products to stock more of in specific stores to maximize profit.

- Business analytics is unlike pure technical data work, it always ties back to business outcomes like revenue or operations.

Source: Analytics.hbs

02 DATA ANALYTICS

- Data analytics focuses on the technical process of examining raw data sets, this includes/entails cleaning, transforming, and modelling these data sets so that ~~we can turn them~~ we can identify patterns or answers to problems
- Data analytics is more technical and can be used in many fields such as healthcare, science, finance and business to provide information businesses use for decision making (business analytics)

Example: A hospital / healthcare provider analyzes patient records to spot trends in treatment outcomes to help reduce readmissions.

Source: fivetran

03 BUSINESS INTELLIGENCE (BI)

- Business Intelligence refers to systems and tools used to collect, organise, and display data through dashboards and reports.
- It mainly helps businesses understand current and previous performance ie, 'What is happening now' or 'what happened'

Example: A sales team uses a BI dashboard in Tableau or PowerBI to track daily revenue and inventory (stock levels)

Source: Tableau

04 DATA-DRIVEN DECISION MAKING

- Data-driven decision making means making choices based on facts and evidence from data instead of relying on your intuition.
- It helps reduce uncertainty and improve outcomes. A positive to this is that companies that use this approach are often more efficient and competitive

Example: An online store reviews customer checkout data before redesigning its website.

05 BIG DATA

- Big data refers to extremely large and complex datasets that grow quickly and come from many sources.
- Since traditional systems struggle to manage large amounts of data, advanced technologies and tools are needed for storage and data processing.
- Businesses use big data to uncover deeper trends and customer behaviour

Example: Social Media Companies analyse billions of likes, posts, and videos daily

06 DESCRIPTIVE ANALYTICS

- Descriptive analytics focuses on summarising data to show what happened, it uses reports, graphs, and summaries to identify trends or performance results.
- It is usually the first step of analytics

Example: A report showing the company that sales decreased by 10% last quarter.

Source: Insightsoftware

07 DIAGNOSTIC ANALYTICS

- Diagnostic analytics looks deeper into data to explain why something happened. It searches for root causes, relationships, or factors built behind a particular result.
- It helps businesses solve problems more effectively

e.g A ~~kear~~ company discovers that sales dropped because of stock shortages in one region.

08 ~~Predictive Analytics~~ PREDICTIVE ANALYTICS

- Uses past data, models and machine learning to estimate (predict) what may happen in the future.
- It helps business prepare for risks or opportunities, however predictions are not guaranteed

Example: A bank predicts ~~discovers sales dropped because of stock shortage in one region~~ which customers are more likely to miss loan payments

Source: Online hbs

09 ~~Prescriptive Analytics~~ PRESCRIPTIVE ANALYTICS

- Prescriptive analytics recommends the best actions to take based on prediction and data. It often uses optimization models, simulations, or AI
- This is one of the most advanced forms of analytics

Example: A delivery company predicts delays and recommends better routes for drivers

10. DATA LITERACY

- Data literacy is the ability to read, understand, interpret and communicate data effectively.
- It allows people to ask better questions and avoid misusing statistics.
- It is an important skill in modern workplaces

Example: ~~Customer names, purchase dates, and sales amount in Excel~~ A manager correctly interprets a dashboard results before making decisions

02 DATA, MEASUREMENT & REPORTING

11 STRUCTURED DATA

- Structured data is data stored in an organised format, ie. rows and columns
- It is easy to search, analyse and manage using databases or spreadsheets

E.g Customer names, purchase dates, and sales amounts in excel

12. Unstructured data

- Unstructured data does not follow a fixed format and includes text, videos, emails, and images.
- It usually requires advanced tools to analyze

Example: Customer reviews on a website

13. DATA WAREHOUSE

- A data warehouse is a central storage system where data from different departments is combined for reporting and analysis.
- It is designed for decision making rather than daily operations.

Example: Combining sales, finance and HR data into one system

14. DATA MINING

- This is the process of exploring large data sets to discover hidden patterns or relationships.
- It often uses algorithms to find useful insights

Example: A supermarket finds that ~~the~~ customers often buy bread and milk together.

15. DATA VISUALISATION

- Data visualisation means presenting data using charts, graphs, or dashboards to make information easier to understand.
- Good visuals help people spot trends quickly

E.g A line graph showing monthly website visits

16. DASHBOARD

A dashboard is a visualisation tool that displays important metrics in one place, often in real time, it helps managers track performance quickly

e.g A marketing dashboard showing clicks, leads and ad spend

17. METRIC

- A metric is any measurable value used to track activity or performance. Not all metrics are strategic.

Example: Number of website visitors

18. KEY PERFORMANCE INDICATORS (KPI)

- KPI is a metric directly linked to business goals and success, it is used to measure progress towards objectives.

Example: Monthly revenue growth

Source: Datapad

19. BENCHMARK

- A benchmark is a standard used to compare business peformance against competitors, industry averages or past results.

Example: Comparing your customer satisfaction score with industry averages.

20. SEGMENTATION

- Segmentation means dividing customers or data into groups with similar characteristics.
- This allows more targeted strategies.

Example: Grouping customers into high-spending and low-spending categories

03 CUSTOMER & SALES KPI's

21. CONVERSION RATE

- Conversion rate is the percentage of people who complete a desired action out of the total number who had the opportunity

Example: 500 purchases from 10 000 website visitors = 5%

22. CUSTOMER ACQUISITION COST (CAC)

- CAC measures how much it costs to gain one new customer through sales and marketing efforts

Examples: Spending R100 000 to gain 1000 customers = R100 CAC

23. CUSTOMER LIFETIME VALUE (CLV)

- CLV estimates the total profit a business expects from a customer over the full relationship

Example: A customer generates R5000 profit over the years

Source: Online.hbs

24 CHURN RATE

- Churn rate is the percentage of customers who stop using a service during a certain period.
- lower churn is better

Example: Losing 5% of subscribers monthly Source: Investopedia

25. CUSTOMER RETENTION RATE

- Retention rate shows the percentage of customers a business keeps overtime. High retention often means loyal customers

Example: Retention 85% of customer Annually.

26. NET PROMOTER SCORE (NPS)

- NPS measures customer loyalty by asking how likely customers are to recommend the company

Example: A company with a score of 70 has strong loyalty

27 AVERAGE ORDER VALUE (AOV)

- AOV is the average amount spent per order or transaction

Example: Increasing average cart value from R500 to R650

28 SALES FUNNEL

- The sales funnel shows the stages customers move through before making a purchase.
- It helps identify where people drop off.

Example: Leads move from ad clicks → sign up → purchase

04 MARKETING & CAMPAIGN ANALYTICS

29 MARKETING CAMPAIGN

- A marketing campaign is a planned set of promotional activities designed to archieve a business goal

Example: A holiday sale promoted through email and social media

30 Post Campaign Analysis

This is the review of camp
future campaigns

Example: ~~Campa~~ checkin

31 INCREMENTAL REV

- Incremental Revo
beyond normal sales

Example:

32. RETURN OF

- ROI measure

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Source: Investopedia

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30 Post Campaign Analysis

This is the review of campaign results after it ends to measure success and improve future campaigns

Example: ~~Campr~~ checking which ad platform produced the most sales.

31 INCREMENTAL REVENUE

- Incremental Revenue is the extra ~~tos~~ income earned because of a campaign or action, begond normal sales levels

Example: A promo campaign brings in an extra R5000-

32. RETURN OF INVESTMENT (ROI)

- ROI measures how profitable an investment was compared to its cost

Example: Spending R10 000 and making R30 000 profit = Strong ROI

33. RETURN ON AD SPEND (ROAS)

ROAS measures revenue earned for every rand spent on advertising

Examples: R5 earned for every R1 spent on ads-

Source: Reportdash

34 CLICKS-THROUGH RATE (CTR)

- CTR is the percentage of people who clicked an ad or link ~~the~~ after seeing it

Example: 2 out of every 100 viewers clicked

35 BOUNCE RATE

Bounce Rate is the percentage of visitors who leave a website after viewing only one page

Example: A 70% bounce rate may suggest poor landing page quality

36 A/B TESTING

A/B testing compares two versions of something to see which performs better

Example: Testing two ad headlines to improve clicks

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37 LEAD GENERATING

- Lead generation is the process of attracting potential customers and collecting their contact details for future leads

Example: Offering a free guide in exchange for an email address.

38 MARKETING ATTRIBUTION

- Marketing attribution identifies which marketing channels contributed to a sale or conversion

Example: Discovering that email marketing influenced a purchase before the final ad click.

Source: Business.adobe